# FORCING OF CLIMATE

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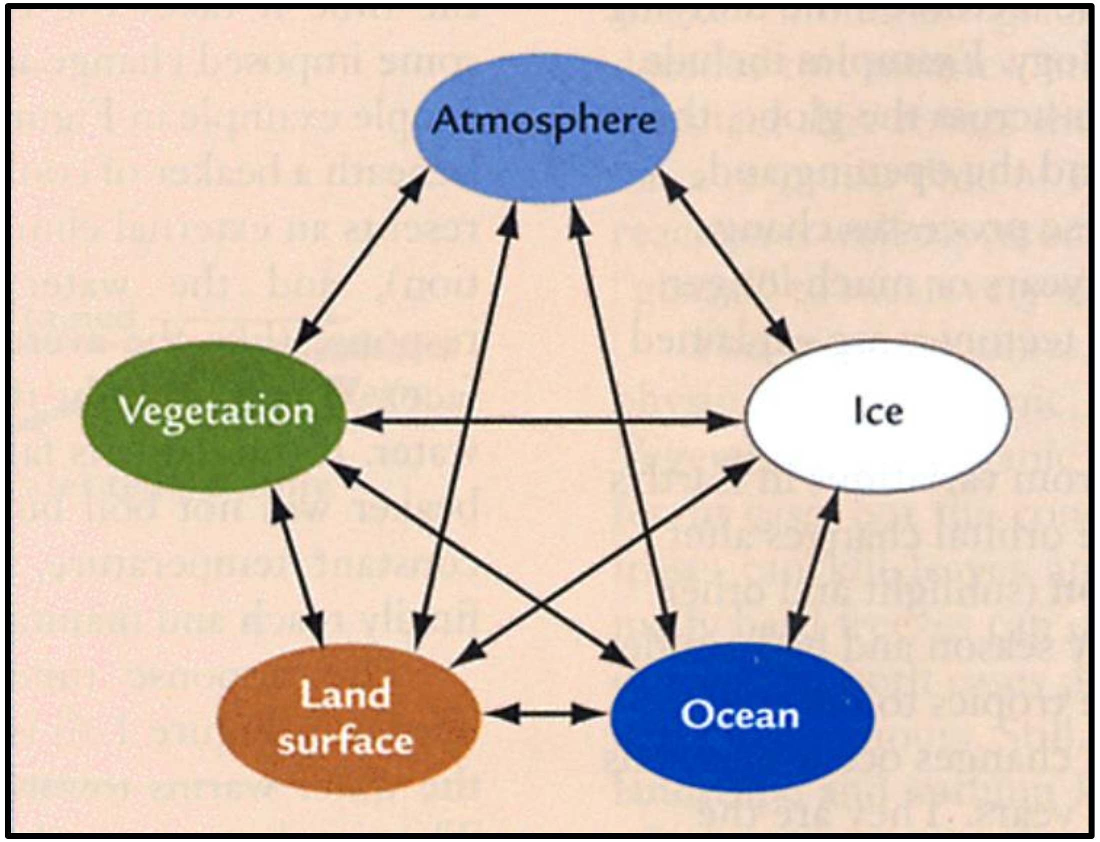
# CLIMATE FORCING

**CLIMATE SYSTEM**
**(Interactions)**




**Lecture 1:** The climate system evolves in time under the influence of its own internal dynamics (*Natural Climate Variability*)

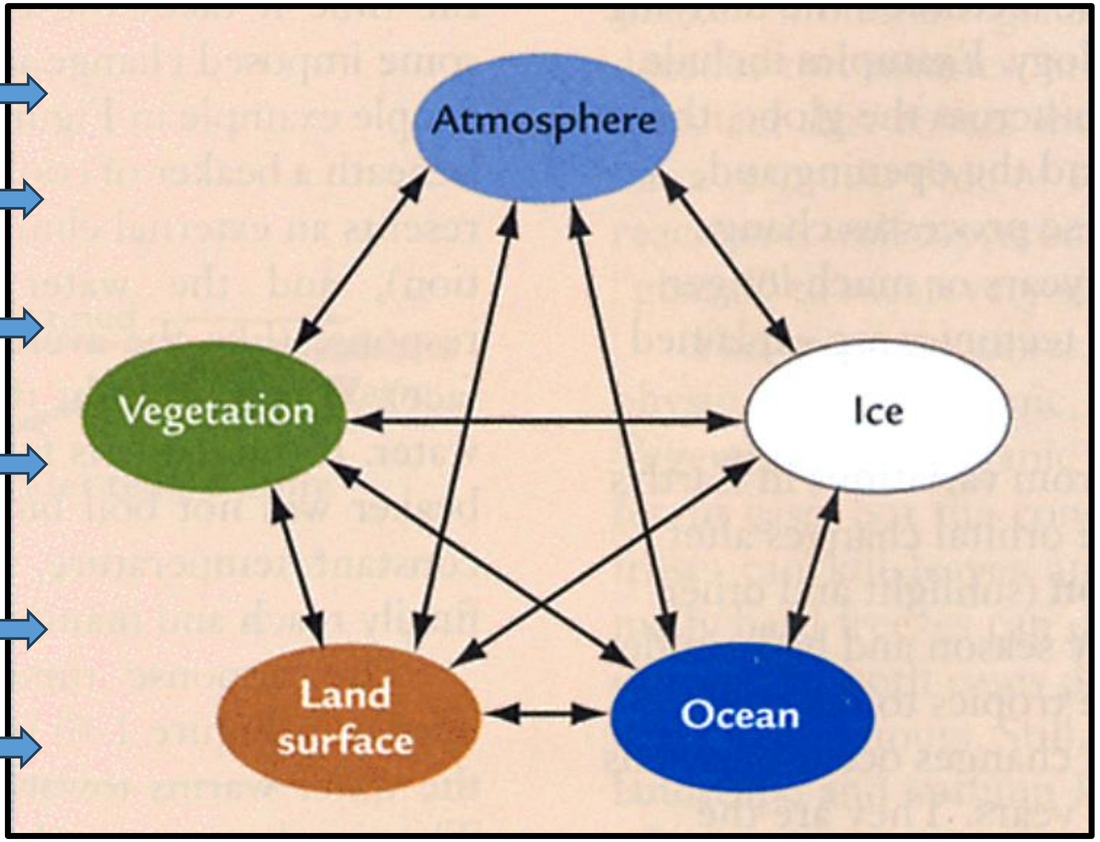
# CLIMATE FORCING

## Factors that influence Earth's climate

**CAUSES**
**(External forcing)**

- Changes in plate tectonics
- Changes in Earth's orbit
- Changes in solar activity
- Ocean – Atmosphere interactions
- Volcanic eruptions
- Anthropogenic activities

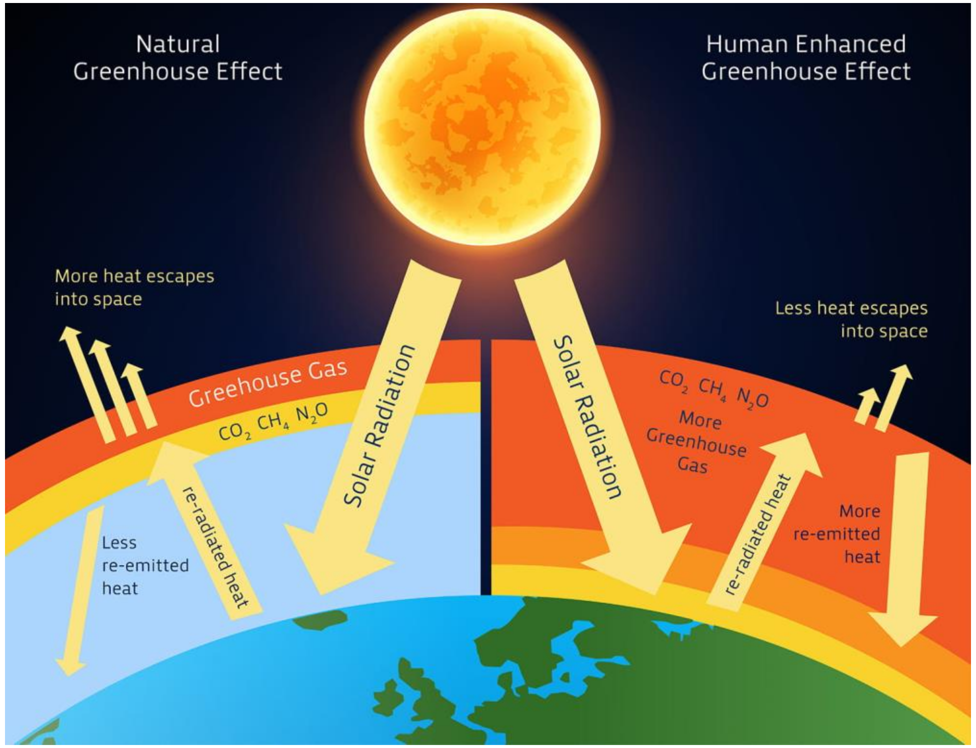
**CLIMATE SYSTEM**
**(Interactions)**




**Climate Forcing:** an imposed change to the Earth's energy balance, either by natural or human-induced factors, that can lead to a change in the Earth's climate

# CLIMATE FORCING

## Factors that influence Earth's climate




### The Greenhouse Effect

***Incoming Solar Radiation***

- The sun is the primary energy source for Earth's climate
- The sun's radiation reaches Earth, and some of it is reflected back into space

***Earth's Surface Absorbs Heat***

- The remaining solar energy is absorbed by the Earth's surface (land and oceans), warming it

***Heat Radiates from Earth***

- The Earth's surface then emits heat back into space in the form of infrared radiation

***Greenhouse Gases Trap Heat***

- Greenhouse gases in the atmosphere (like carbon dioxide, methane, and water vapor) absorb some of this outgoing infrared radiation

***Warming Effect***

- By trapping heat, these gases prevent it from escaping into space, leading to a warmer Earth's surface and lower atmosphere

# CLIMATE FORCING

## Factors that influence Earth’s climate

### Ocean – atmosphere interactions

***Energy and Heat Exchange***

- The ocean is a heat sink ~ stores more heat than the atmosphere
- Heat transfer through evaporation, influencing temperature and humidity
- Impacts weather patterns ~ formation of storms and precipitation

***Water and Moisture Exchange***

- The ocean → major source of atmospheric moisture through evaporation
- Water vapor through the atmosphere ~ cloud formation and precipitation

***Gas Exchange***

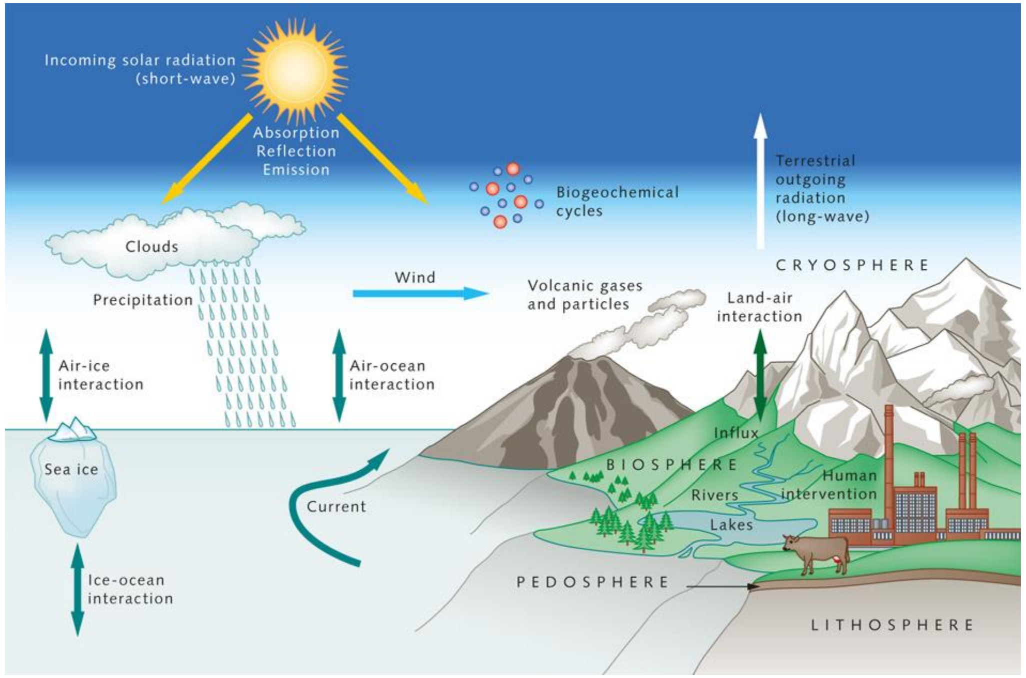
- The ocean and atmosphere exchange various gases ~ $CO_2$ and $O_2$
- Affects ocean chemistry, marine ecosystems, and atmos composition

***Momentum Exchange***

- Winds affect the ocean surface, driving currents and affecting sea level ~ sea level rise and upwelling
- The ocean's surface roughness and temp can influence wind patterns

# CLIMATE FORCING

## Factors that influence Earth's climate




### More Natural Factors

**Variations in Solar Radiation and Earth's Orbit**

- Variations in solar output and Earth's orbital parameters can alter the amount of solar energy received, impacting global temperatures

**Volcanic Activity**

- Volcanic eruptions release aerosols and gases into the atmosphere, which can reflect sunlight and lead to cooling effects, albeit usually short-lived
    - ➢ Tiny particles, either liquid or solid, suspended in the air

**Greenhouse Gases**

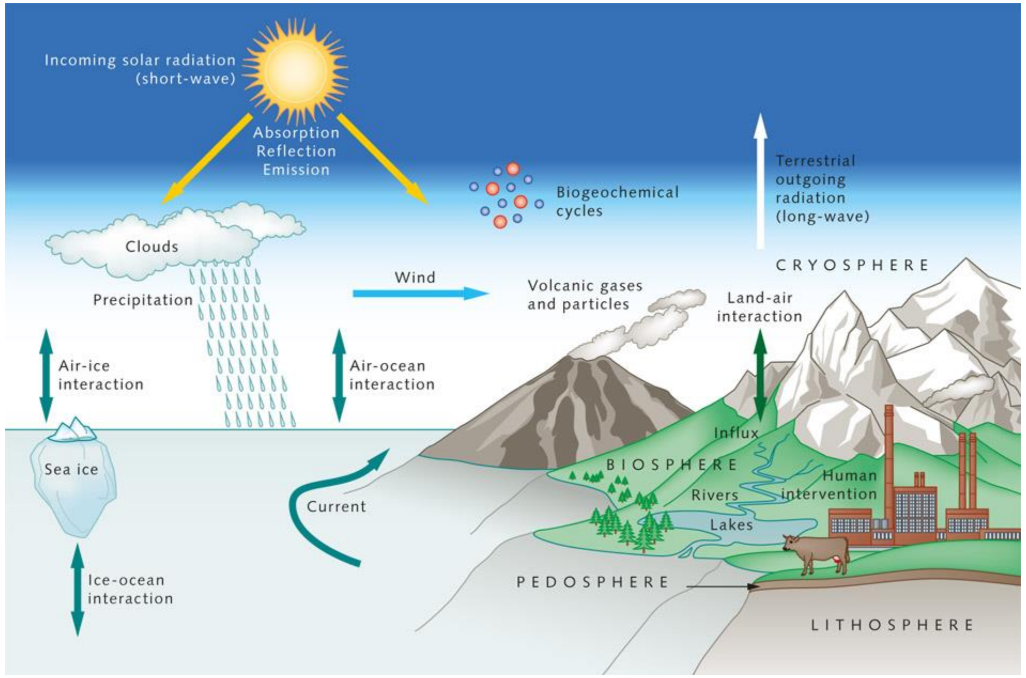
- Natural greenhouse gases trap heat in the atmosphere, contributing to the greenhouse effect and keeping the planet warm enough for life

**Earth's Orbit**

- Changes in Earth's orbit (Milankovitch cycles) alter the distribution of solar radiation across the planet, triggering long-term climate shifts

# CLIMATE FORCING

## Factors that influence Earth's climate




### Human Factors

**Burning Fossil Fuels**

- Releases large amounts of $CO_2$ into the atmosphere, enhancing the greenhouse effect and contributing to global warming

**Deforestation**

- Trees absorb $CO_2$ from the atmosphere. Deforestation reduces this carbon sink, leading to a higher concentration of greenhouse gases and potentially warmer temperatures

**Agriculture**

- Agricultural practices, particularly the use of fertilizers and livestock farming, release greenhouse gases like methane and nitrous oxide

**Industrial Processes**

- Industrial activities, such as manufacturing and electricity generation, also contribute to greenhouse gas emissions

# CLIMATE FORCING

## Factors that influence Earth's climate

### CAUSES (External forcing)

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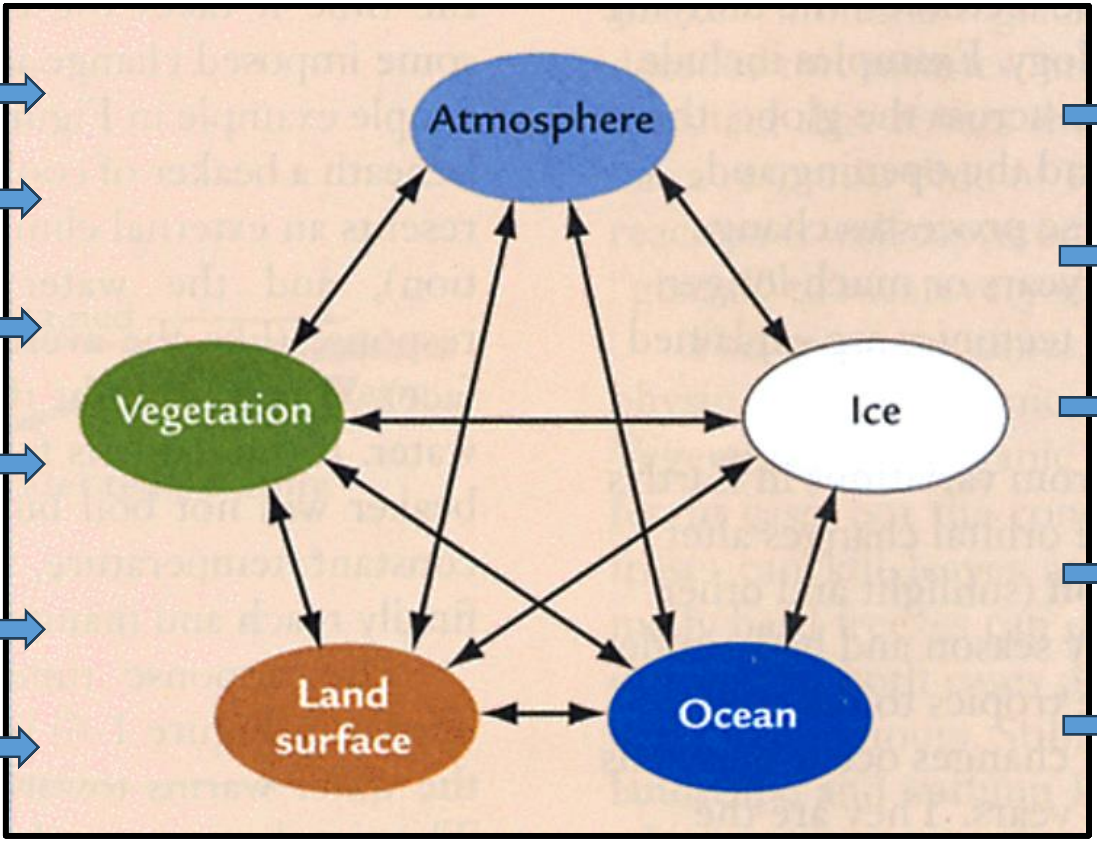
### CLIMATE SYSTEM (Interactions)




### CLIMATE VARIATIONS (Internal responses)

- Changes in atmosphere
- Changes in ice
- Changes in vegetation
- Changes in ocean
- Changes in land surface

# CLIMATE FORCING

## Factors that influence Earth's climate

**CAUSES (External forcing)**

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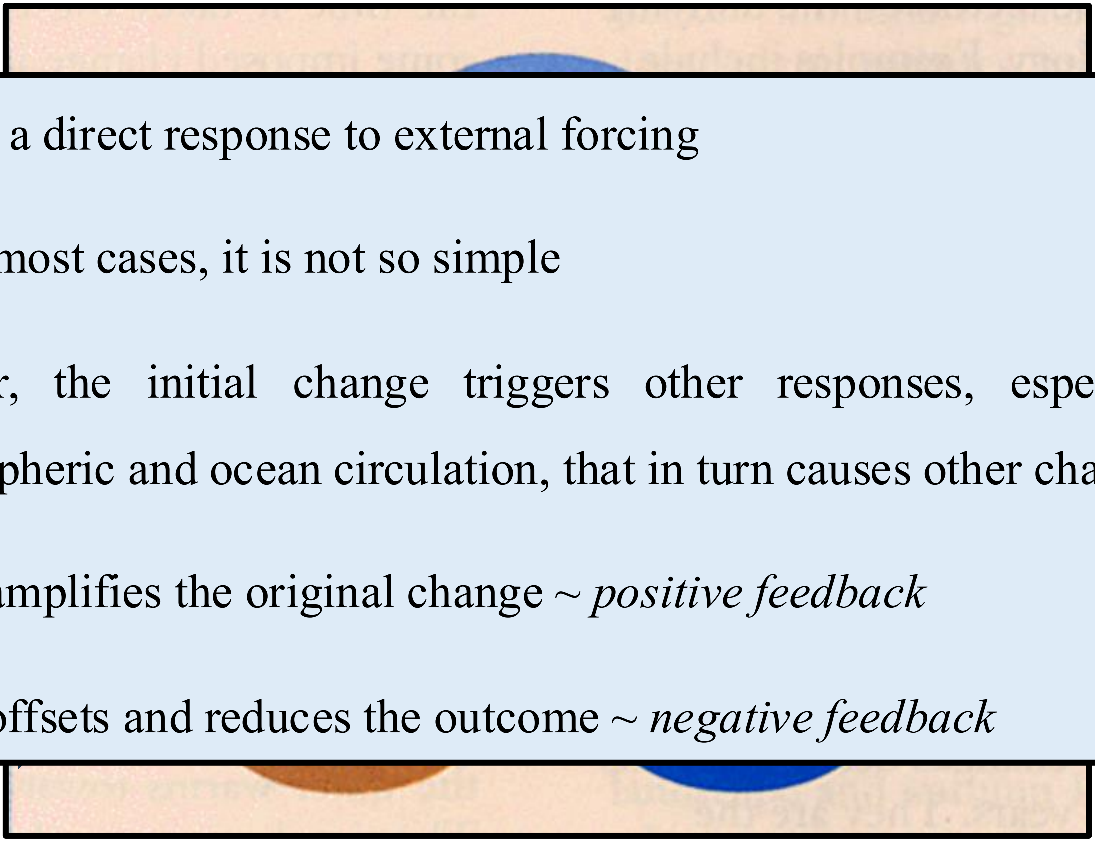
**CLIMATE SYSTEM (Interactions)**



**CLIMATE VARIATIONS (Internal responses)**

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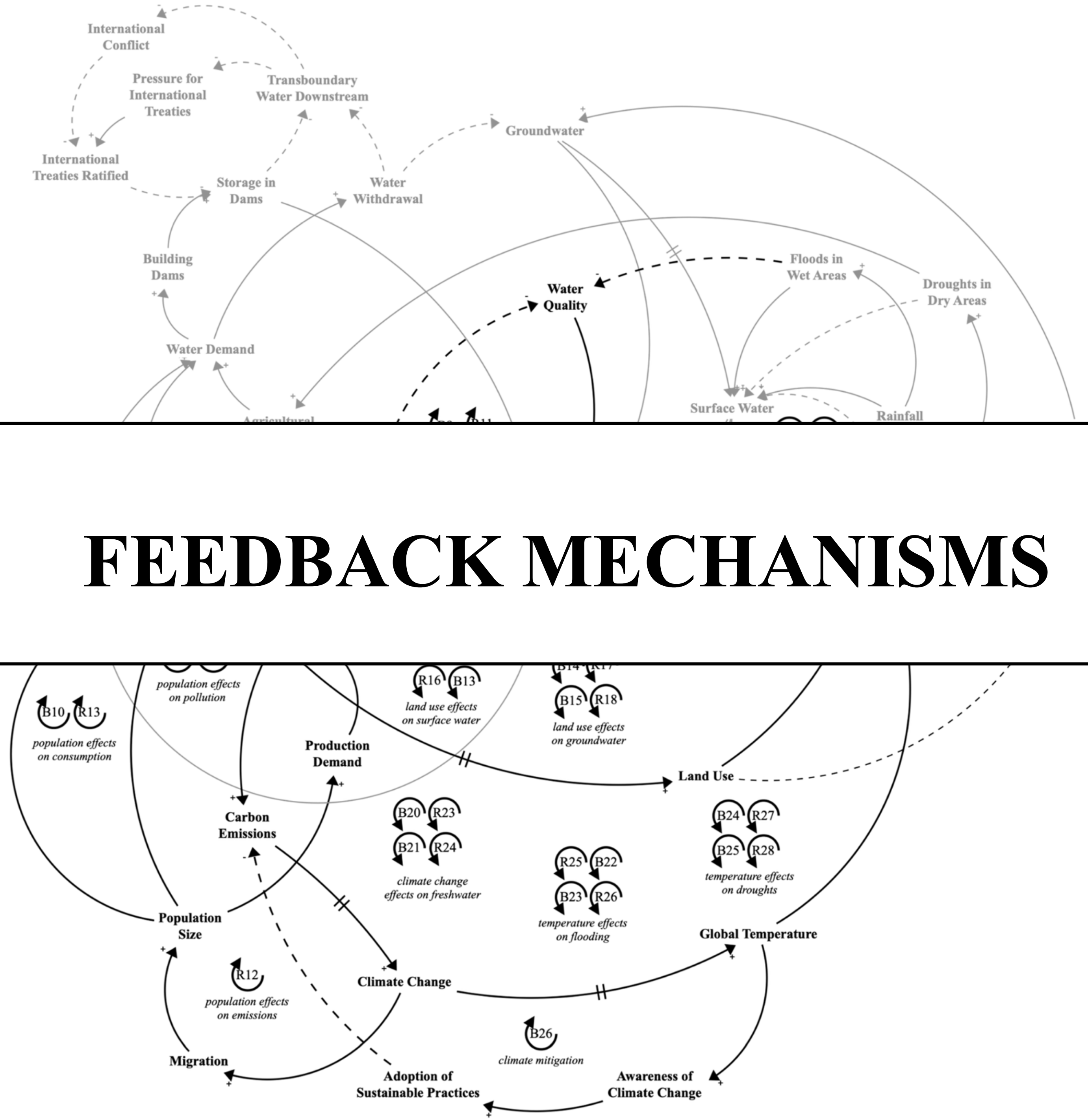
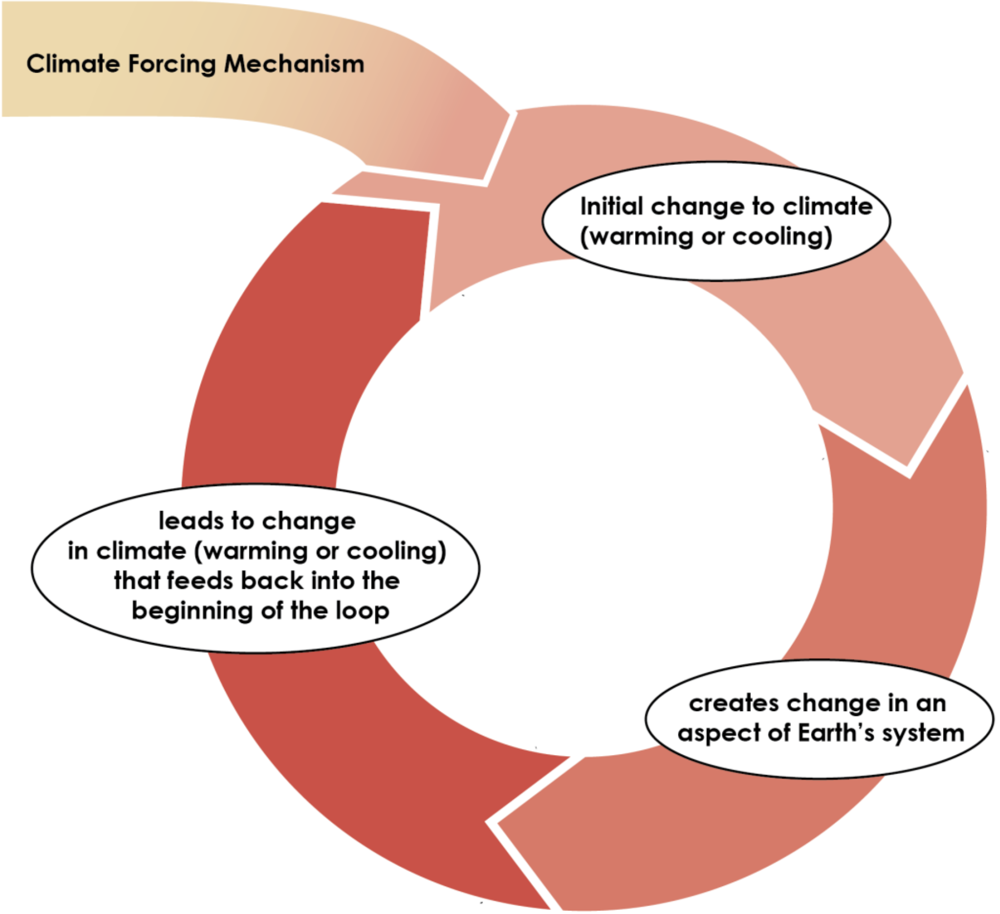
- Overall, there is a direct response to external forcing
- But in many or most cases, it is not so simple
  - Rather, the initial change triggers other responses, especially in the atmospheric and ocean circulation, that in turn causes other changes to occur
- If the response amplifies the original change ~ *positive feedback*
- If the response offsets and reduces the outcome ~ *negative feedback*

# FEEDBACK MECHANISMS



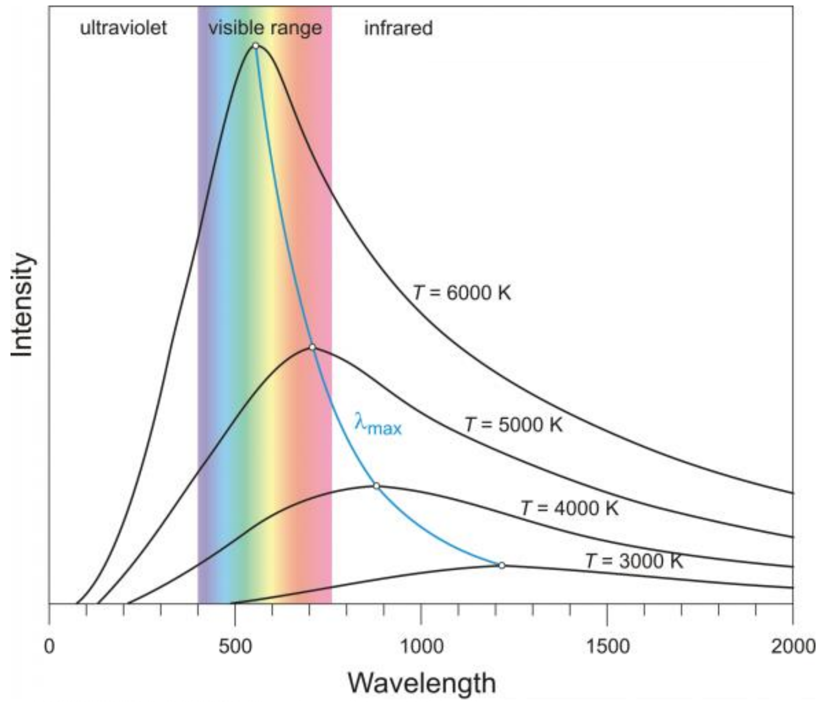
# FEEDBACK MECHANISMS

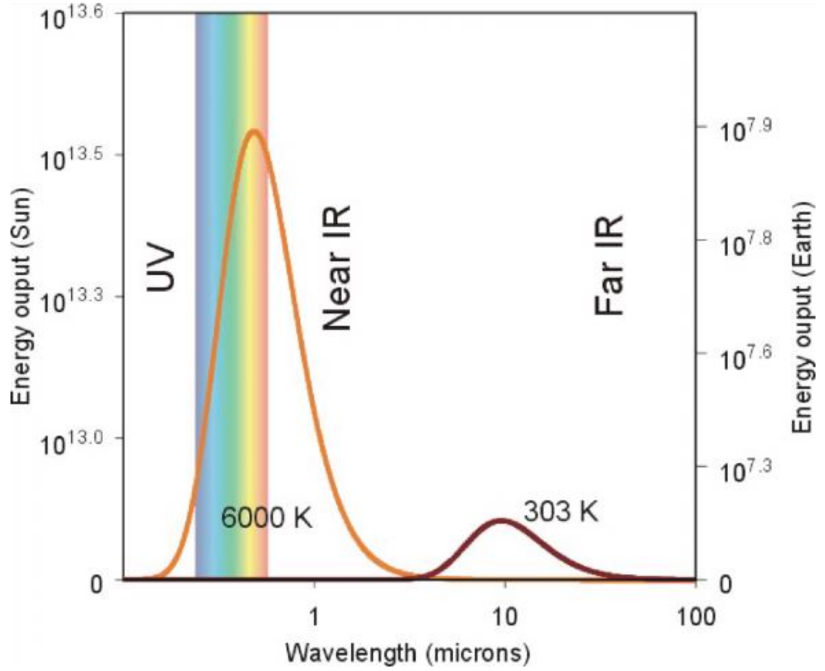



- A process "internal" to the climate system that changes as a function of climate, and in turn affects (feeds back to) climate

| | | |
|---|---|---|
| *examples* | - changes in snow/ice coverage | (+) |
| | - changes in cloud distributions | (?) |
| | - water vapour in the atmosphere | (+) |
| | - vegetation distribution changes | (+) |
| | - changes in carbon sinks | (+) |

(+) - change is reinforced, (-) – change is "damped"




# FEEDBACK MECHANISMS

## Planck Radiation Feedback

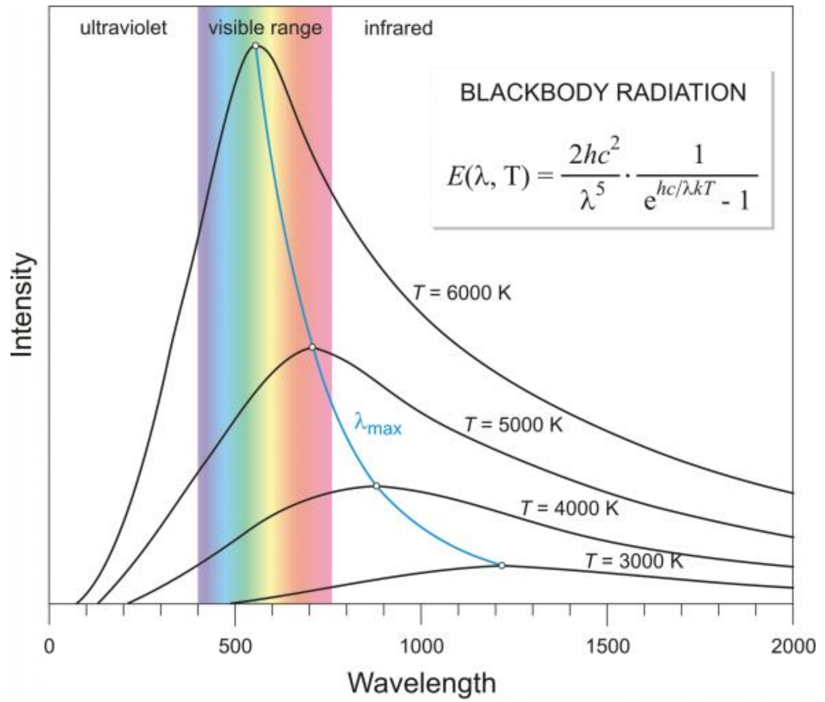
- Describes how changes in Earth's outgoing longwave radiation (OLR) influence its temperature in response to warming
- The outgoing longwave radiation emitted by the Earth depends on surface temperature, due to the Stefan-Boltzmann Law
- Stefan-Boltzmann law ~ hotter objects emit more radiation

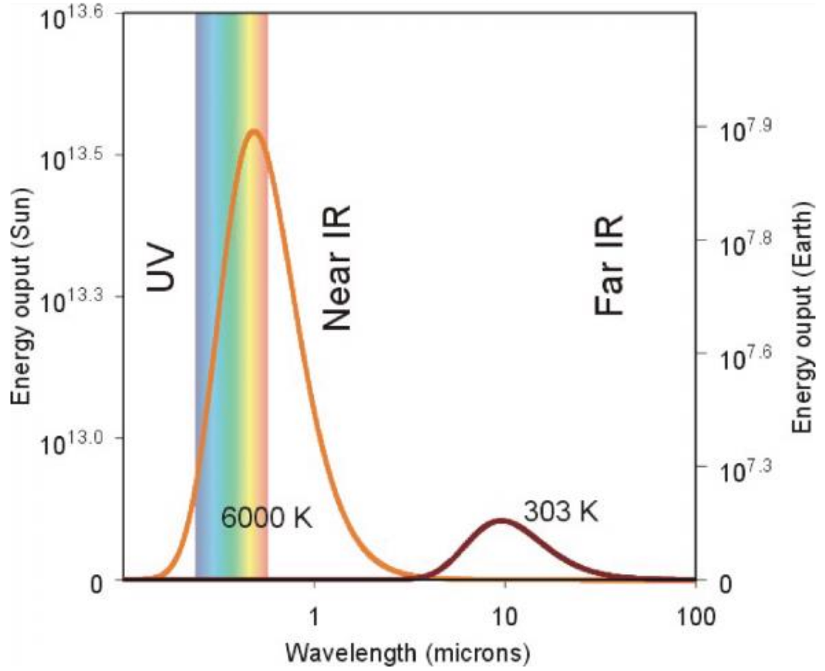
$$E = \sigma T^4$$

E = radiation emitted in W/m²

σ= 5.67 x 10⁻⁸ W/m² * K *sec

T = temperate (K ← *Kelvin degree*)



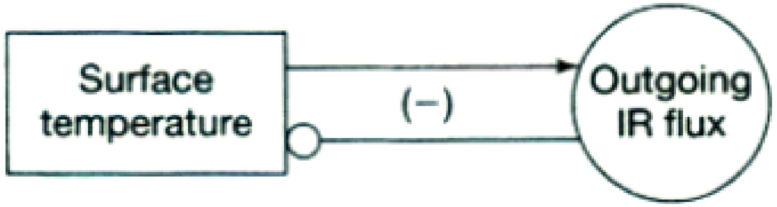

# FEEDBACK MECHANISMS

## Planck Radiation Feedback

- If warming occurs, then temperatures rise
- The ultimate rise in temperature in the absence of feedbacks is determined by the amount of heat, and the mass and specific heat of the body
- But all bodies radiate, and a pure black body radiates with the fourth power of absolute temperature
  - Black body: theoretical perfect absorber and emitter of radiation
- Therefore, as the body heats up, it radiates more and loses some of the heat

$$E = \sigma T^4$$

- E = radiation emitted in W/m$^2$
- $\sigma$= 5.67 x $10^{-8}$ W/m$^2$ * K *sec
- T = temperate (K ← *Kelvin degree*)



# FEEDBACK MECHANISMS

## Ice – Albedo Feedback

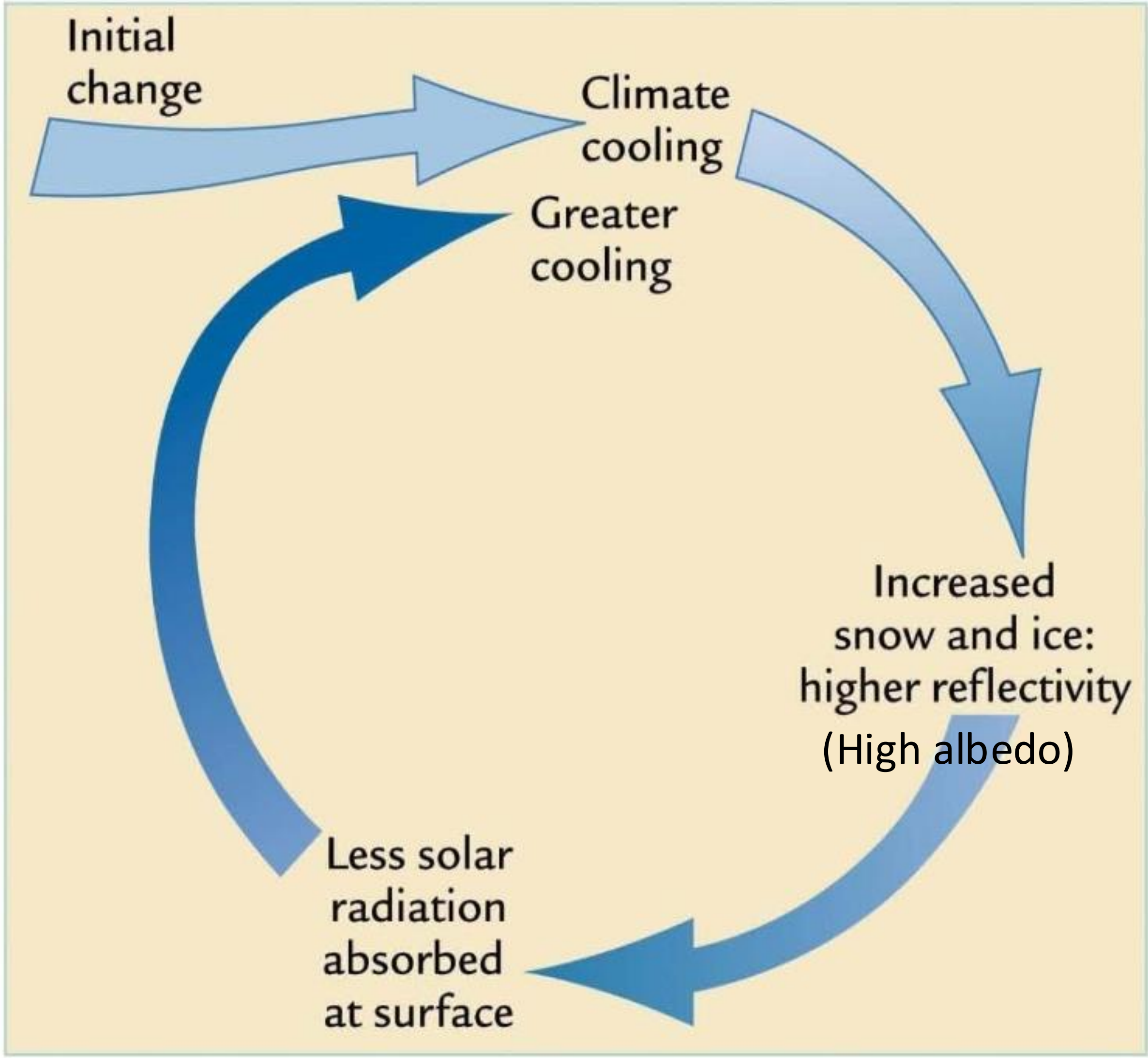


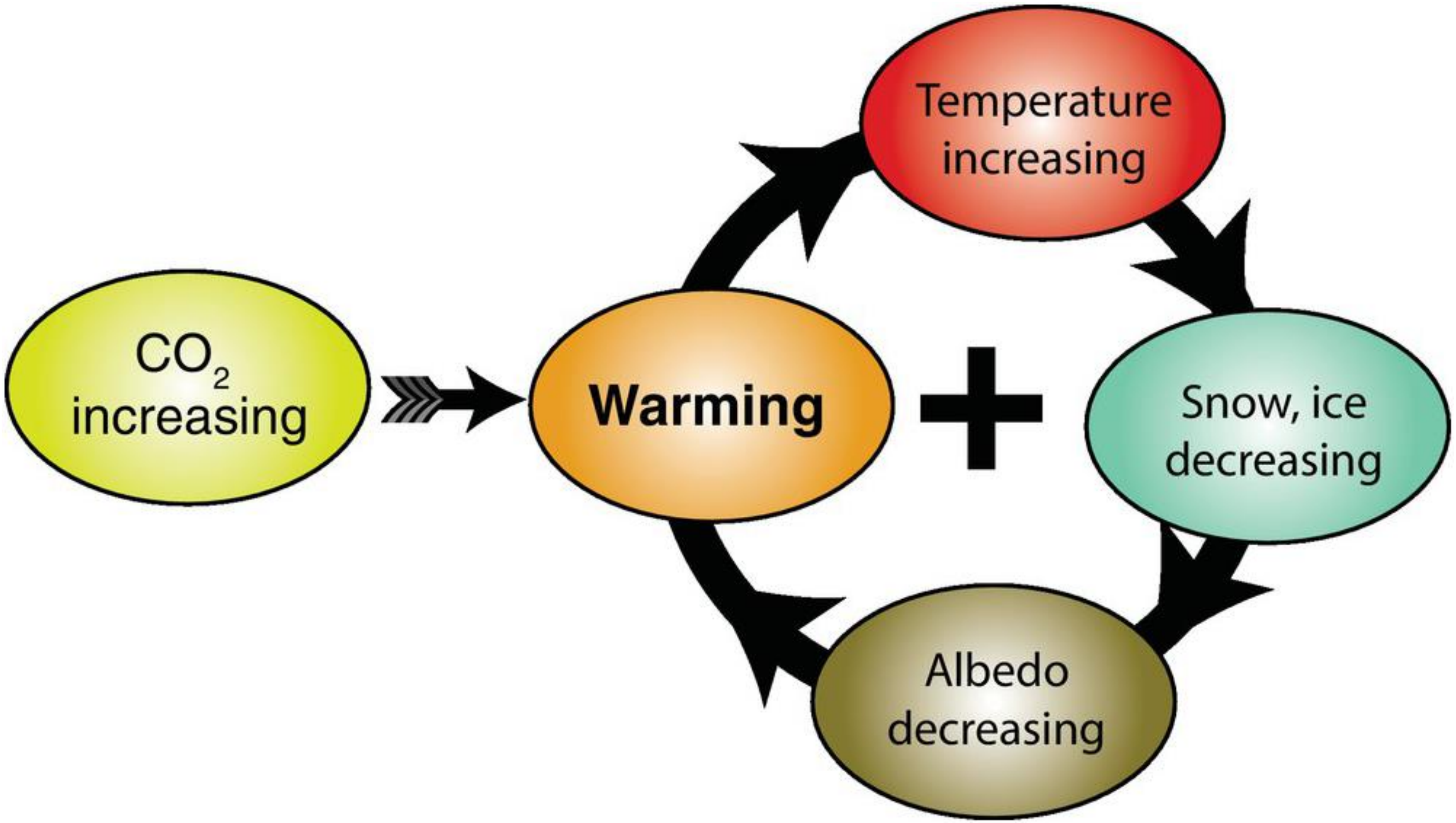

**TABLE 2-1 Average Albedo Range of Earth's Surfaces**

| Surface | Albedo range (percent) |
|---|---|
| Fresh snow or ice | 60–90% |
| Old, melting snow | 40–70 |
| Clouds | 40–90 |
| Desert sand | 30–50 |
| Soil | 5–30 |
| Tundra | 15–35 |
| Grasslands | 18–25 |
| Forest | 5–20 |
| Water | 5–10 |

*Adapted from W. D. Sellers,* Physical Climatology *(Chicago: University of Chicago Press, 1965), and from R. G. Barry and R. J. Chorley,* Atmosphere, Weather, and Climate, *4th ed. (New York: Methuen, 1982).*

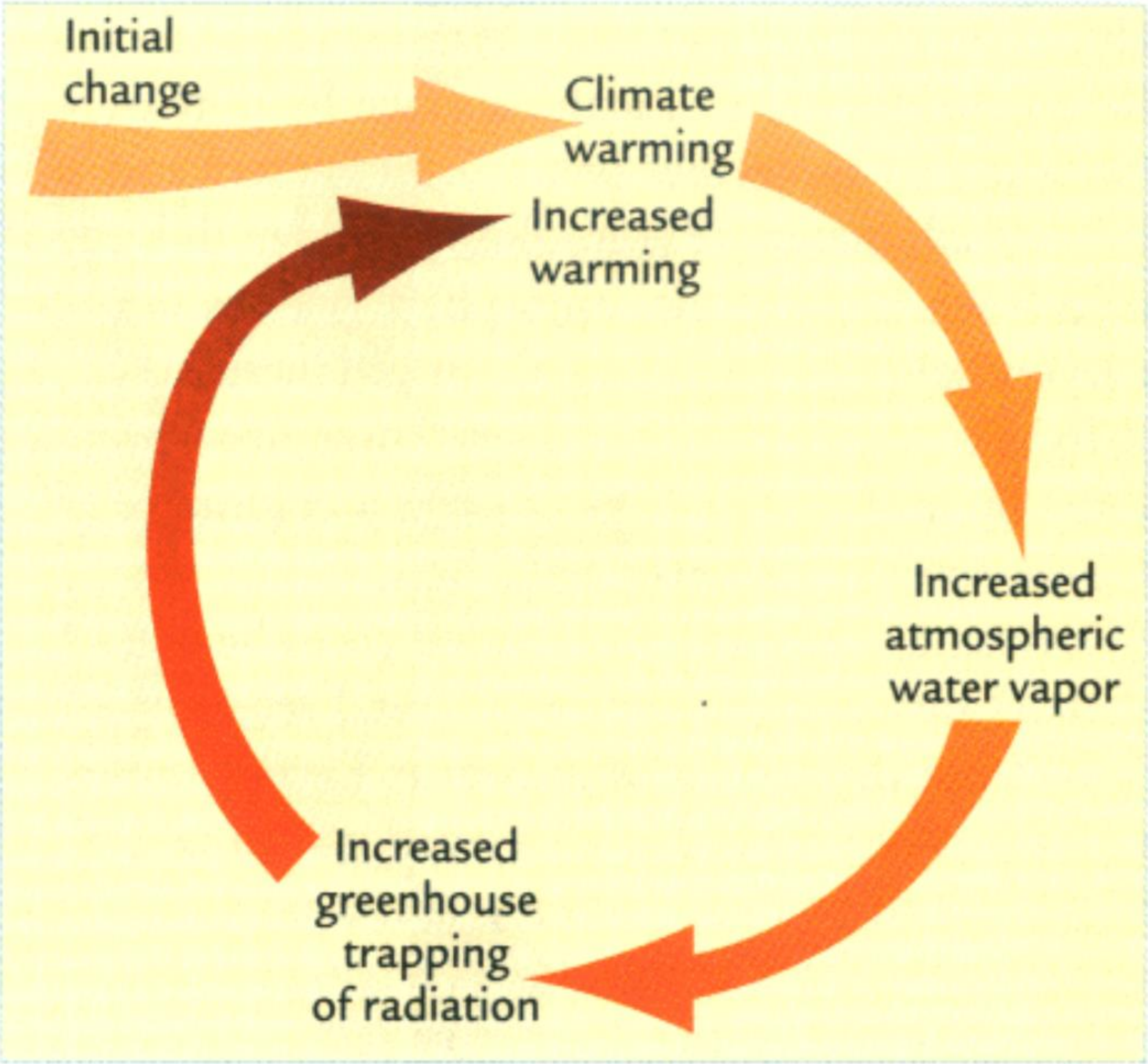
# FEEDBACK MECHANISMS

## Ice – Albedo Feedback



# FEEDBACK MECHANISMS

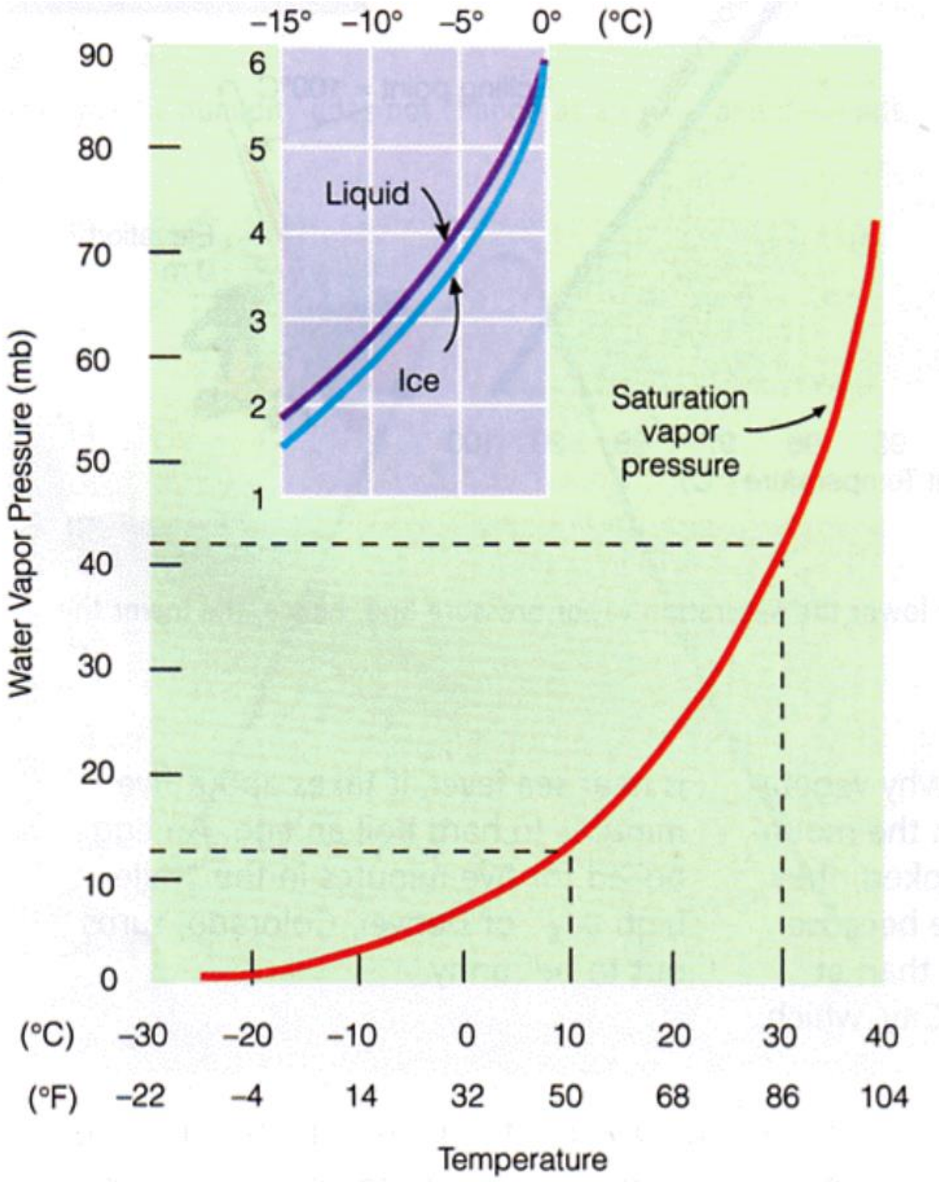
## Water Vapor Feedback




Remember: Water vapor is a greenhouse gas

# FEEDBACK MECHANISMS

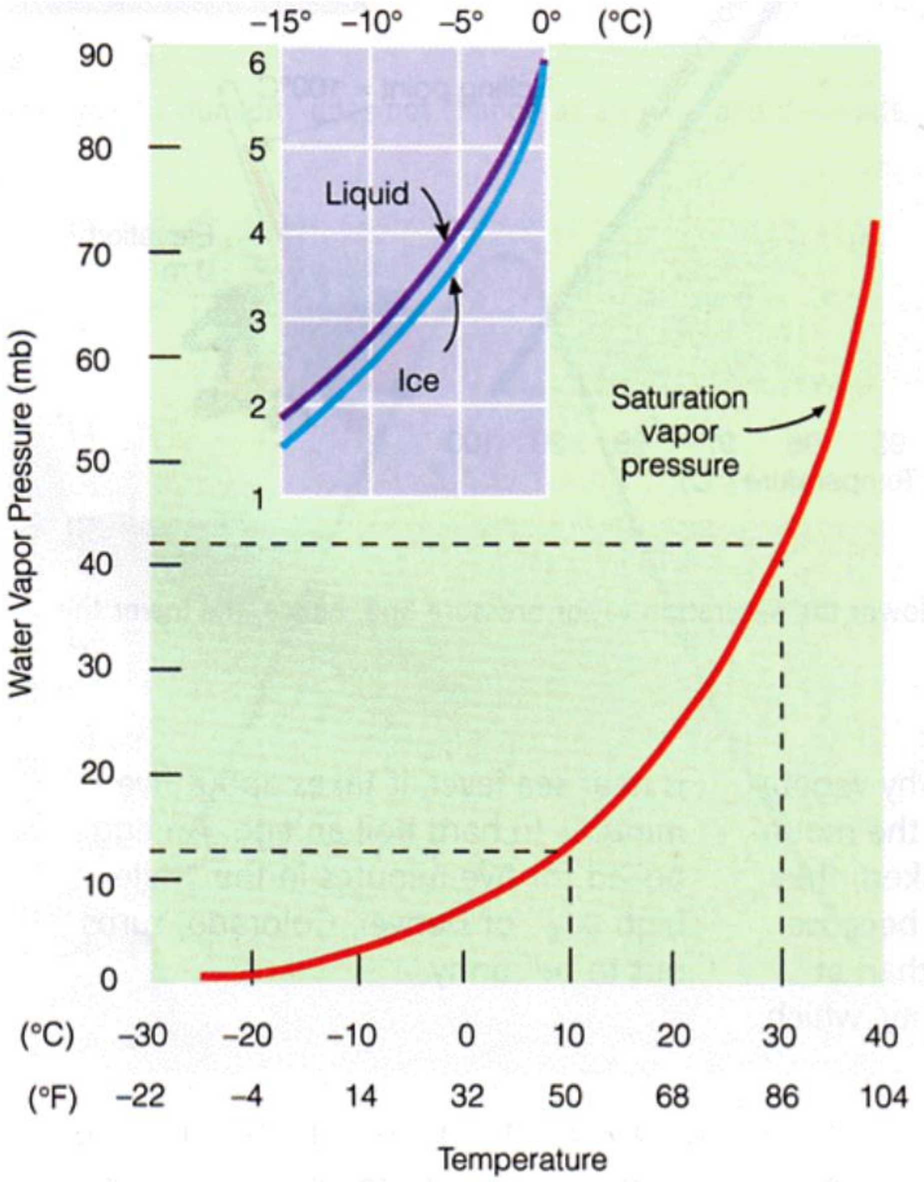
## Water Vapor Feedback




- Water vapor exponentially increases as temperature increases
- The water-holding capacity of the atmosphere increases with temperatures
  - ➢ Based on the Clausius–Clapeyron relationship

# FEEDBACK MECHANISMS

## Water Vapor Feedback




**The Clausius–Clapeyron relationship**

- Describes how saturation vapor pressure (the maximum amount of water vapor air can hold at a given temperature) increases with temperature
- Saturation vapor pressure depends primarily on the air temperature in the following way:

$$\frac{de_s}{dT} = \frac{L}{T(\alpha_v - \alpha_l)} \qquad e_s \cong 6.11 \cdot \exp\left\{\frac{L}{R_v}\left(\frac{1}{273} - \frac{1}{T}\right)\right\}$$

***Saturation Vapor Pressure (es):*** The pressure exerted by water vapor at saturation

***Temperature (T):*** Usually measured in Kelvin (K)

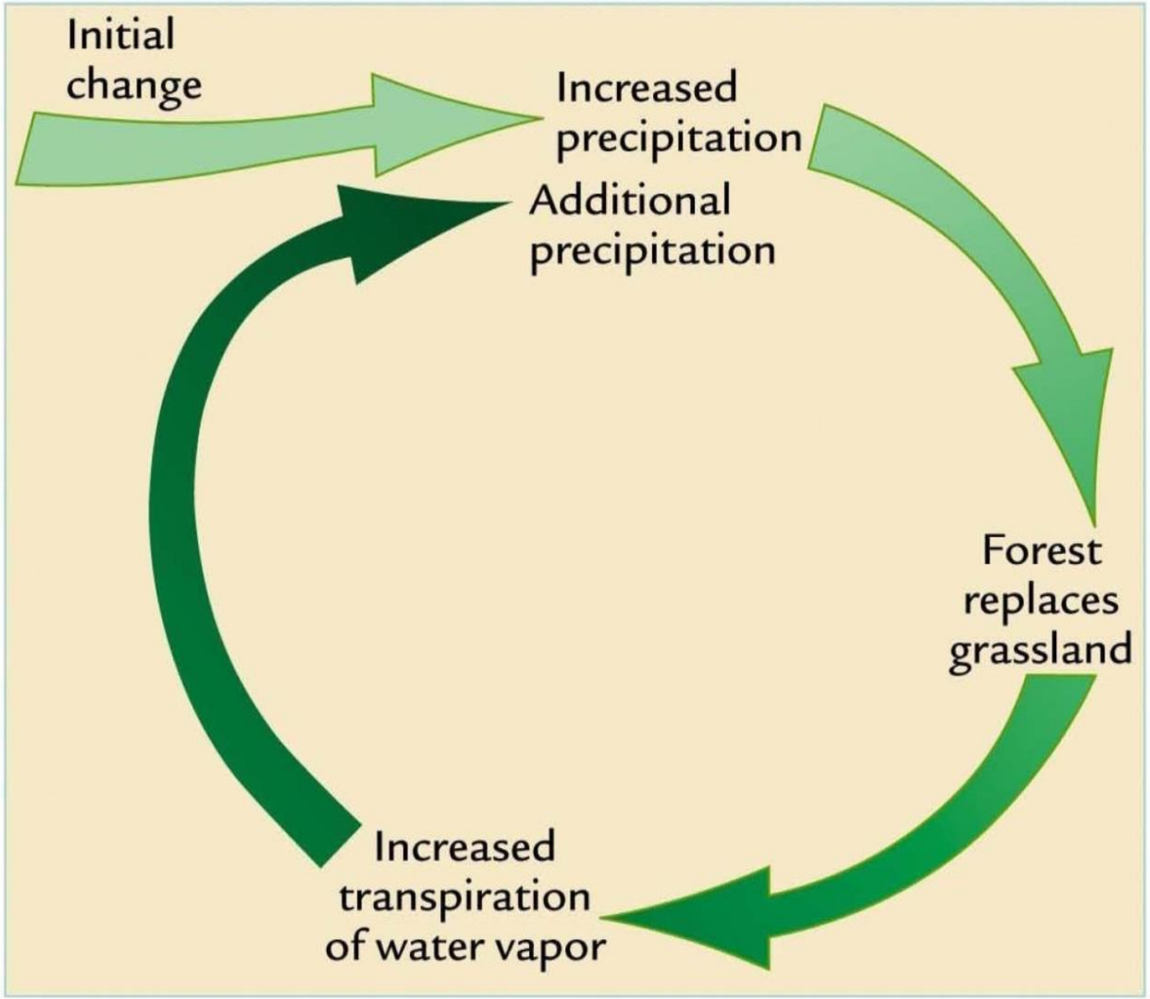
***Latent Heat of Vaporization (Lv):*** The energy required to evaporate water

***Gas Constant for Water Vapor (Rv):*** constant related to properties of water vapor

***α (or sometimes ΔV):*** specific volume of vapor and liquid

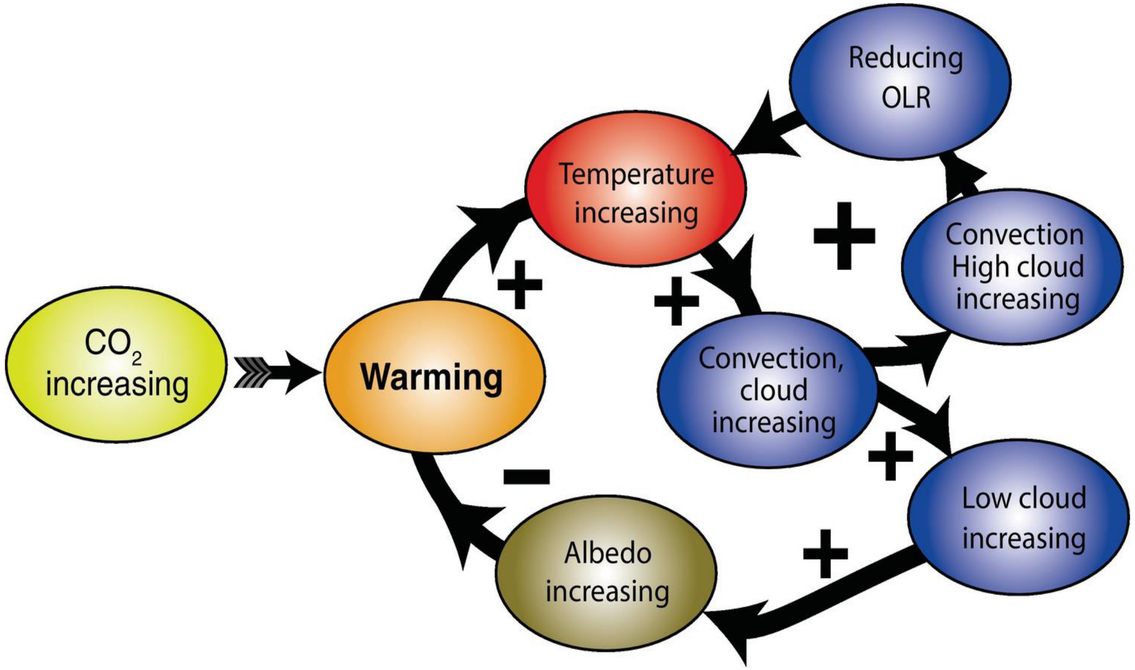
# FEEDBACK MECHANISMS

## Vegetation – Precipitation Feedback



# FEEDBACK MECHANISMS

## Cloud Feedback




**Negative Feedback**

- Low, thick clouds reflect incoming solar radiation back to space
  - ➢ High albedo
- Cools the climate

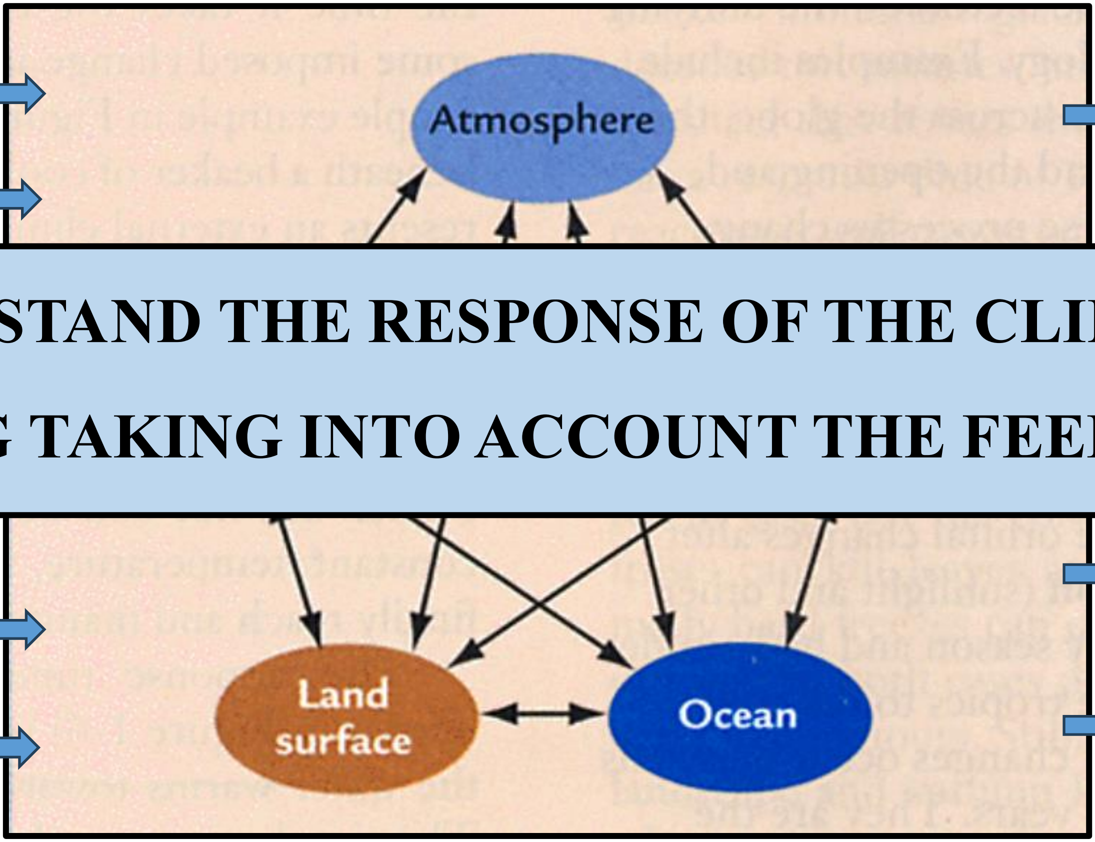
**Positive Feedback**

- High, thin clouds have cold cloud tops that radiate less to space
- Allow incoming solar radiation to pass through
  - ➢ Low albedo
- Also trap some of the outgoing infrared radiation (longwave)
- Warms the climate

**CAUSES**
**(External forcing)**

- Changes in plate tectonics
- Changes in Earth's orbit
- Volcanic eruption
- Anthropogenic activities

**CLIMATE SYSTEM**
**(Interactions)**

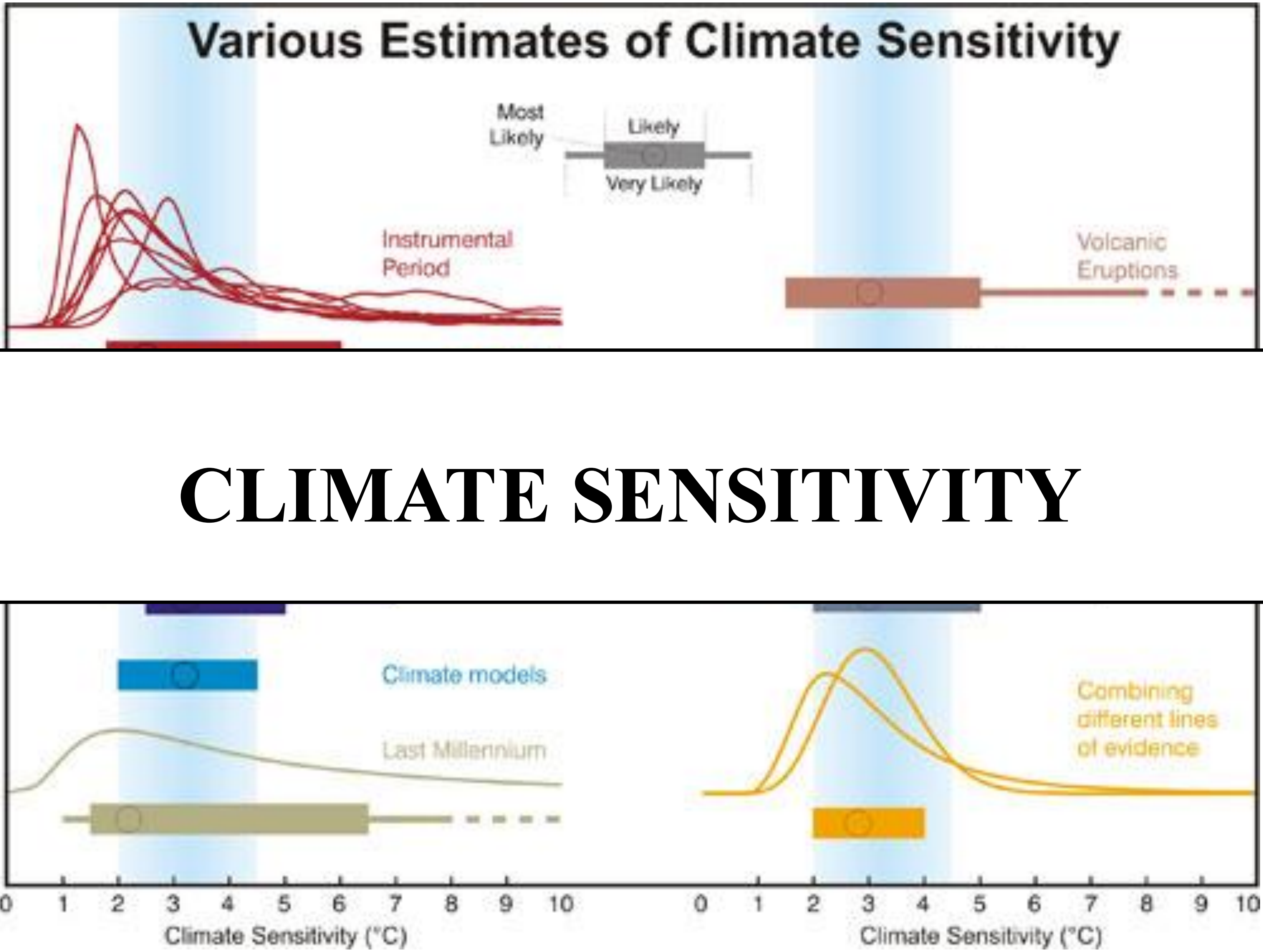



**CLIMATE VARIATIONS**
**(Internal responses)**

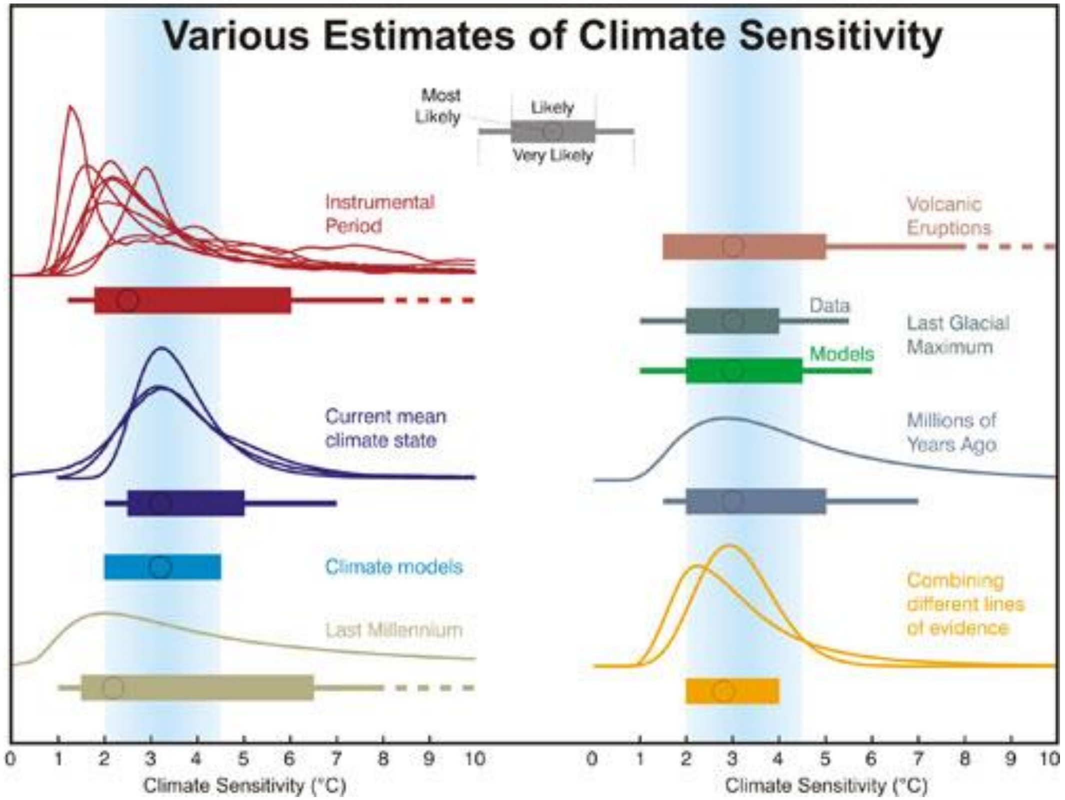
- Changes in atmosphere
- Changes in ocean
- Changes in land surface

**HOW DO WE UNDERSTAND THE RESPONSE OF THE CLIMATE SYSTEM TO A GIVEN FORCING TAKING INTO ACCOUNT THE FEEDBACK LOOPS?**

# CLIMATE SENSITIVITY



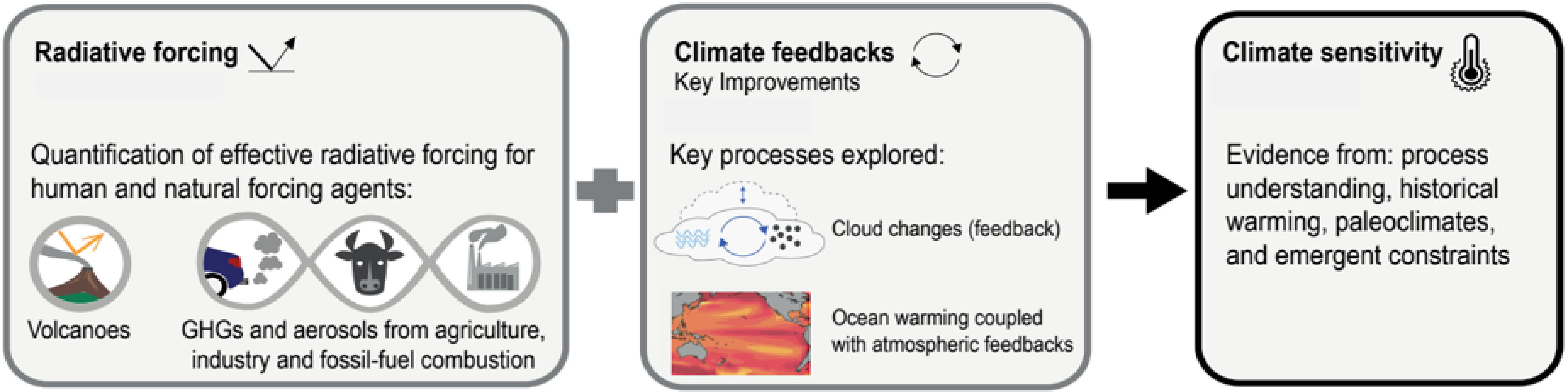
# CLIMATE SENSITIVITY




- A term used to characterize the response of the climate system (or components) to a given forcing
- Idealized value depicting the sensitivity of climate models to a perturbation
- The degree to which Earth's climate warms when the atmospheric carbon dioxide concentration doubles

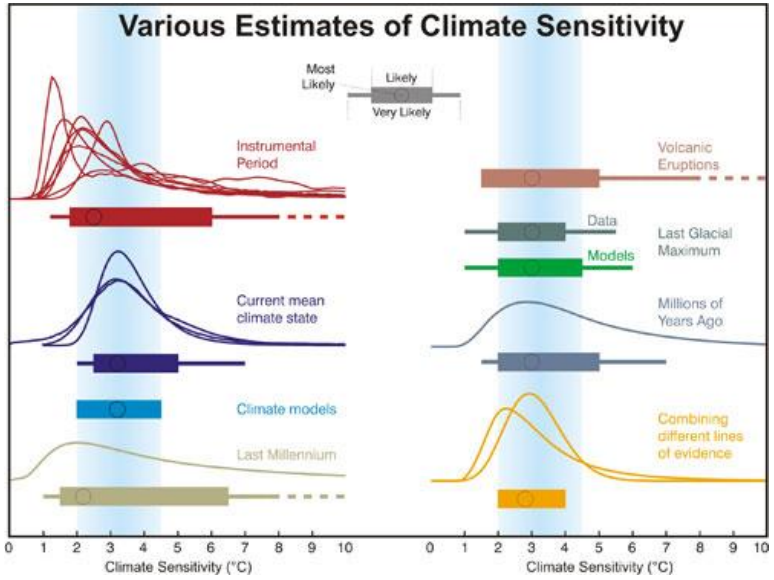
# CLIMATE SENSITIVITY

Earth's energy budget is influenced by:




- More specific (formal) definitions:
  - *Equilibrium climate sensitivity (ECS)*: the equilibrium (steady state) change in the surface temperature following a doubling of the atmospheric carbon dioxide ($CO_2$) concentration from pre-industrial conditions.
  - *Transient climate response (TCR)*: the surface temperature response for the hypothetical scenario in which atmospheric carbon dioxide ($CO_2$) increases at 1%/yr from pre-industrial to the time of a doubling of atmospheric $CO_2$ concentration (year 70).

# CLIMATE SENSITIVITY




**Estimating Climate Sensitivity**

***Analyzing historical climate data***

- Examining past climate changes and their potential drivers

***Using climate models***

- Simulating the climate system to predict how it will respond to different scenarios of greenhouse gas emissions

***Considering climate feedbacks***

- Analyzing the complex interactions within the climate system that amplify or dampen the initial warming

# CLIMATE SENSITIVITY

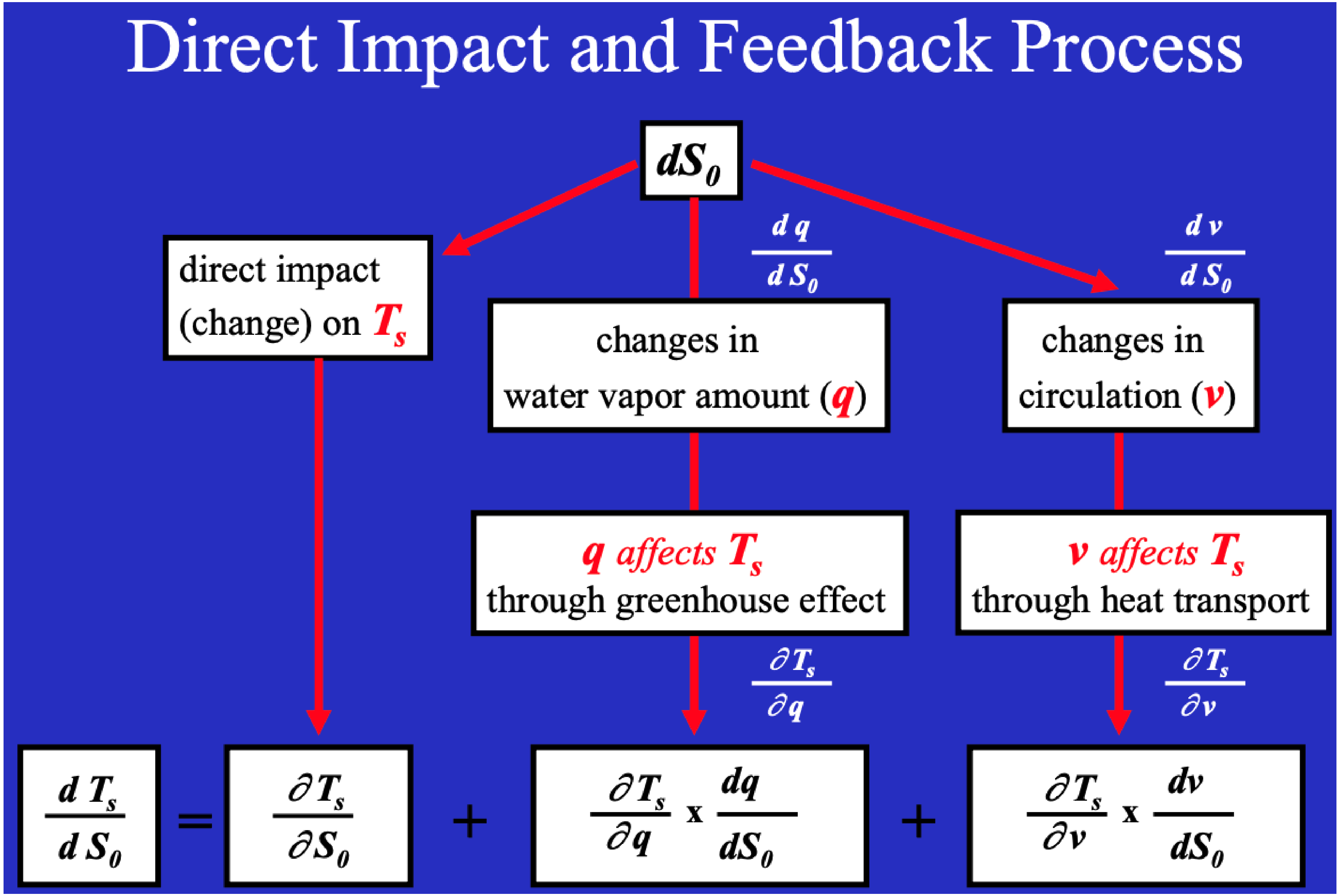
direct impact

$$\frac{dT_s}{dS_0} = \frac{\partial T_s}{\partial S_0} + \sum_{j=1}^{N} \frac{\partial T_s}{\partial y_j}\frac{dy_j}{dS_0}$$

- Climate Sensitivity: the relationship between the measure of forcing and the magnitude of the climate change response.
- Feedback Mechanism: a process that changes the sensitivity of the climate response.
- Climate Forcing (***S₀***): Climate forcing is a change to the climate system that can be expected to change the climate.

ESS200
Prof. Jin-Yi Yu

# CLIMATE SENSITIVITY



**THANK YOU**